Natalie Leung
014790932
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Data 260 Homework 8

Personal Configuration and Domain

SID4	0932
PORT_BASE	8032
PREFIX	s0932
SEED	0932
VERIFY_SEED	260932
DOMAIN_ID	4 (open-source package vulnerabilities)

Hardware: nvdia geforce 4070, msi cpu
Local model used: ollama qwen3:8b
Tagged commit hash repo with most updated codes (submission on canvas only):
https://github.com/n-k-leung/data260-0932

Part 1 - Backend Service Request/Response over Kafka

This homework builds on HW7. Instead of calling your domain logic directly inside one process, you'll split it into a client and a worker that talk only through Kafka messages. This lets you see problems that a single process hides, such as duplicate, lost, or out-of-order messages.

I. Operations

This part must be implemented for your primary domain entity. Replace ENTITY with the actual entity name used by your assigned domain. For example, if your entity is Course, CREATE_ENTITY becomes CREATE_COURSE and entity_id becomes course_id. Replace <PREFIX> with the prefix calculated in Section 0.

Worker (consumer.py)

```python
def process_request(operation, data):
    if not isinstance(data, dict):
        return {"success": False, "error": "data must be a JSON object"}

    if operation == "CREATE_VULNERABILITY":
        error = validate_vulnerability(data)
        if error:
            return {"success": False, "error": error}

        vulnerability_id = str(uuid.uuid4())
        vulnerabilities[vulnerability_id] = {
            "vulnerability_id": vulnerability_id,
            "package_name": data["package_name"].strip(),
            "severity": data["severity"],
            "vul_code": data["vul_code"].strip(),
            "vendor_id": data["vendor_id"],
        }
        return {
            "success": True,
            "vulnerability_id": vulnerability_id,
            "message": "Vulnerability created",
        }
    elif operation == "GET_VULNERABILITY":
        vulnerability_id = data.get("vulnerability_id")
        if not vulnerability_id:
            return {"success": False, "error": "vulnerability_id is required"}
        if vulnerability_id not in vulnerabilities:
            return {"success": False, "error": "Vulnerability not found"}
        return {"success": True, "vulnerability": vulnerabilities[vulnerability_id]}
    else:
        return {"success": False, "error": "Unknown operation"}
```

```python
+ async def main():
+     consumer = AIOKafkaConsumer(
+         REQUEST_TOPIC,
+         bootstrap_servers=KAFKA_BOOTSTRAP_SERVERS,
+         group_id="s0932-vulnerability-worker-group",
+         auto_offset_reset="earliest",
+         enable_auto_commit=True,
+     )
+     producer = AIOKafkaProducer(bootstrap_servers=KAFKA_BOOTSTRAP_SERVERS)
+
+     await consumer.start()
+     await producer.start()
+     log("Consumer is ready and listening...")
+     try:
+         async for msg in consumer:
+             request = json.loads(msg.value.decode())
+             correlation_id = request.get("correlation_id")
+             reply_to = request.get("reply_to")
+             operation = request.get("operation")
+             data = request.get("data")
+             log(f"WORKER RECEIVE from topic={msg.topic} partition={msg.partition} offset={msg.offset}")
+             log(f"WORKER RECEIVE message={json.dumps(request)}")
+             log(f"WORKER PROCESS operation={operation} correlation_id={correlation_id}")
+             result = process_request(operation, data)
+             response = {"correlation_id": correlation_id}
+             response.update(result)
+             metadata = await producer.send_and_wait(reply_to, json.dumps(response).encode())
+             log(f"WORKER REPLY to topic={metadata.topic} partition={metadata.partition} offset={metadata.offset}")
+             log(f"WORKER REPLY message={json.dumps(response)}")
+     finally:
+         await consumer.stop()
+         await producer.stop()
```

II. Kafka Requirements

Kafka client (kafka_rpc.py)

```python
class KafkaRPC:
    async def consume_responses(self):
        async for msg in self.consumer:
            response = json.loads(msg.value.decode())
            correlation_id = response.get("correlation_id")

            log(f"[API] RECEIVE from topic={msg.topic} partition={msg.partition} offset={msg.offset}")
            log(f"[API] RECEIVE message={json.dumps(response)}")

            if correlation_id in self.pending:
                future = self.pending.pop(correlation_id)
                if not future.done():
                    future.set_result(response)
            else:
                log(f"[API] No pending request for correlation_id={correlation_id}")

    async def make_request(self, request):
        correlation_id = request.get("correlation_id") or str(uuid.uuid4())

        message = {
            "correlation_id": correlation_id,
            "reply_to": RESPONSE_TOPIC,
            "operation": request.get("operation"),
            "data": request.get("data", {}),
        }

        future = asyncio.get_running_loop().create_future()
        self.pending[correlation_id] = future

        metadata = await self.producer.send_and_wait(REQUEST_TOPIC, json.dumps(message).encode())
        log(f"API SEND to topic={metadata.topic} partition={metadata.partition} offset={metadata.offset}")
        log(f"API SEND message={json.dumps(message)}")
        try:
            response = await asyncio.wait_for(future, timeout=TIMEOUT_SECONDS)
            log(f"API RETURN response to caller for correlation_id={correlation_id}")
            return response
        except asyncio.TimeoutError:
            self.pending.pop(correlation_id, None)
            return {
                "correlation_id": correlation_id,
                "success": False,
                "error": "Timed out waiting for the worker response",
            }
```

III. General Requirements

Error handling screenshots



IV. Demonstration

Create a file named homework_demo.py.

homework_demo.py

```python
def send_request(correlation_id, operation, data):
    request = {"correlation_id": correlation_id, "operation": operation, "data": data}
    log(f"CLIENT SEND {operation} -> {REQUEST_TOPIC}")
    log(f"CLIENT SEND message={json.dumps(request)}")

    response = requests.post(API_URL, json=request, timeout=15).json()

    log(f"CLIENT RESPONSE from {RESPONSE_TOPIC} (through FastAPI)")
    log(f"CLIENT RESPONSE message={json.dumps(response)}")
    return response


print("\ncreating vulnerability")

create_correlation_id = str(uuid.uuid4())
log(f"CLIENT created correlation_id={create_correlation_id}")

vulnerability_data = {
    "package_name": "pip",
    "severity": "High",
    "vul_code": "CVE202023337",
    "vendor_id": 1,
}
create_response = send_request(create_correlation_id, "CREATE_VULNERABILITY", vulnerability_data)

if not create_response.get("success"):
    print("cannot create vul failed:", create_response)
    raise SystemExit(1)

vulnerability_id = create_response["vulnerability_id"]
log(f"CLIENT generated vulnerability_id={vulnerability_id}")

print("\nget vulneraiblity")

get_correlation_id = str(uuid.uuid4())
log(f"CLIENT created correlation_id={get_correlation_id}")
get_response = send_request(get_correlation_id, "GET_VULNERABILITY", {"vulnerability_id": vulnerability_id})

print("\n output")
print("retrieved vulnerability:")
print(json.dumps(get_response.get("vulnerability"), indent=2))

create_preserved = create_response.get("correlation_id") == create_correlation_id
get_preserved = get_response.get("correlation_id") == get_correlation_id
print(f"CREATE correlation_id preserved: {create_preserved} ({create_correlation_id})")
print(f"GET correlation_id preserved: {get_preserved} ({get_correlation_id})")
```

API wrapper

reports > hw08 > app > main.py > handle_vulnerability_operation

```python
from contextlib import asynccontextmanager
from fastapi import FastAPI
from app.kafka_rpc import KafkaRPC

rpc = KafkaRPC()

@asynccontextmanager
async def lifespan(app: FastAPI):
    await rpc.start()
    print("FastAPI server started at http://127.0.0.1:8032")
    yield
    await rpc.stop()

app = FastAPI(title="Vulnerability Kafka Demo", lifespan=lifespan)

@app.get("/")
async def home():
    return {"message": "FastAPI service running"}

# Kafka-based endpoint
@app.post("/vuls")
async def handle_vulnerability_operation(payload: dict):
    response = await rpc.make_request(payload)
    return response
```

V. Required Screenshots and Evidence

1. Docker containers running: Run docker ps. The screenshot must show the Kafka service, the worker/consumer service, and the FastAPI service running. Include a worker log showing "Consumer is ready and listening..." and a FastAPI log showing the server started at http://127.0.0.1:<PORT_BASE>.

Containers running

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08> docker ps
CONTAINER ID   IMAGE              COMMAND                  CREATED          STATUS                    PORT
S                                       NAMES
a391af178d83   hw08-api           "uvicorn app.main:ap…"   11 seconds ago   Up 1 second               0.0.
0.0:8032->8032/tcp, [::]:8032->8032/tcp   s0932_api
bcbd659a366c   hw08-worker        "python worker/consu…"   11 seconds ago   Up 1 second
                                        s0932_worker
b65058d28163   apache/kafka:3.9.2 "/__cacert_entrypoin…"   29 minutes ago   Up 29 minutes             0.0.
0.0:9092->9092/tcp, [::]:9092->9092/tcp   kafka
7859a28879e3   postgres:13        "docker-entrypoint.s…"   6 months ago     Up 35 minutes (healthy)   5432
/tcp                                    airflow-postgres-1
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08> docker logs s0932_api

INFO:     Started server process [1]
INFO:     Waiting for application startup.
08:35:08.663 [API] Listening for Kafka responses on topic: s0932_vulnerability_responses
FastAPI server started at http://127.0.0.1:8032
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8032 (Press CTRL+C to quit)
```

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08> docker logs  s0932_wo
rker
08:35:09.566 Consumer is ready and listening...
```

2. CREATE_ENTITY and GET_ENTITY execution: Show the requests and successful responses from homework_demo.py or an API call. The evidence must show the operation, entity data, generated entity_id, success status, and correlation_id.

Output homework-demo.py

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08> python .\homework-dem
o.py

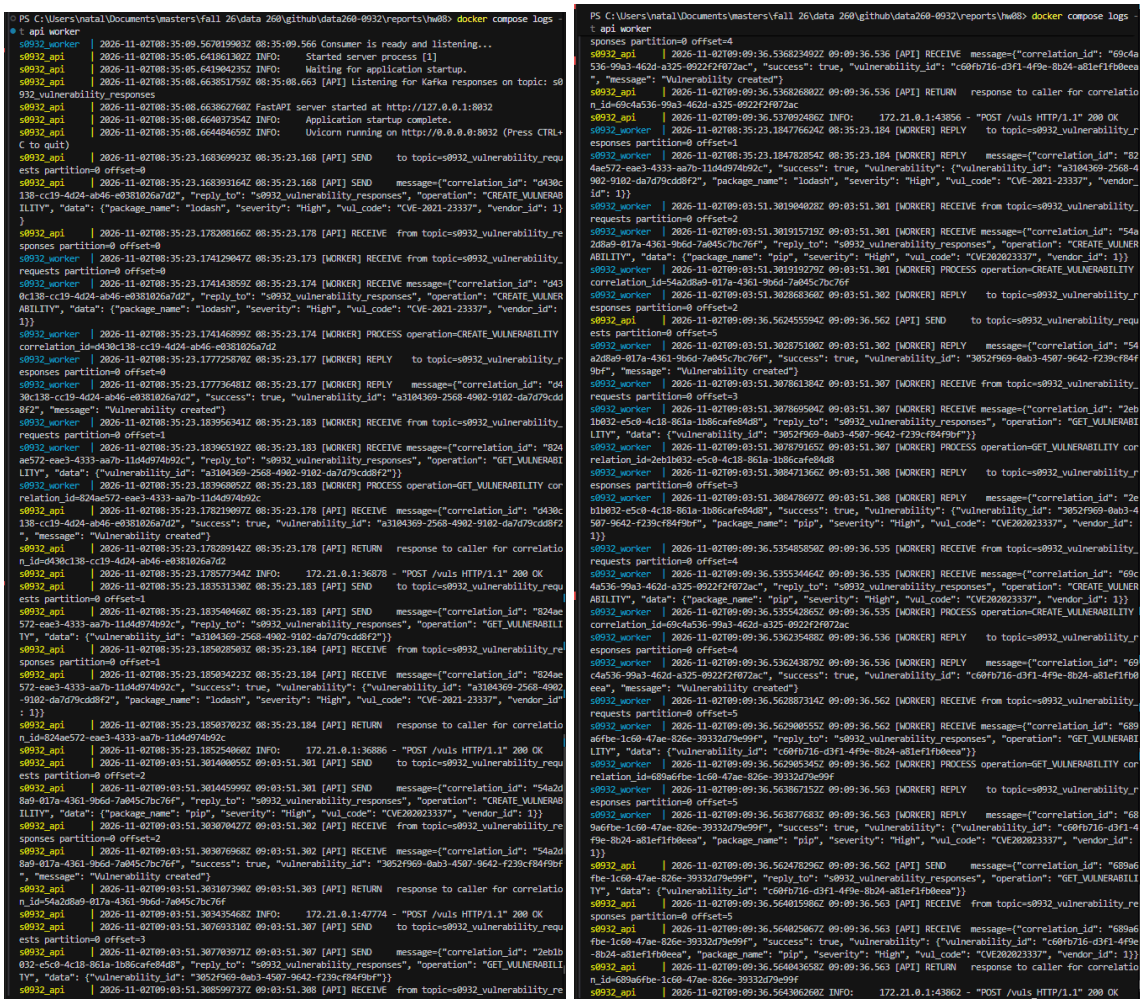
creating vulnerability
01:09:36.526 CLIENT created correlation_id=69c4a536-99a3-462d-a325-0922f2f072ac
01:09:36.526 CLIENT SEND CREATE_VULNERABILITY -> s0932_vulnerability_requests
01:09:36.526 CLIENT SEND message={"correlation_id": "69c4a536-99a3-462d-a325-0922f2f072ac", "operation": "CR
EATE_VULNERABILITY", "data": {"package_name": "pip", "severity": "High", "vul_code": "CVE202023337", "vendor
_id": 1}}
01:09:36.537 CLIENT RESPONSE from s0932_vulnerability_responses (through FastAPI)
01:09:36.537 CLIENT RESPONSE message={"correlation_id": "69c4a536-99a3-462d-a325-0922f2f072ac", "success": t
rue, "vulnerability_id": "c60fb716-d3f1-4f9e-8b24-a81ef1fb0eea", "message": "Vulnerability created"}
01:09:36.537 CLIENT generated vulnerability_id=c60fb716-d3f1-4f9e-8b24-a81ef1fb0eea

get vulneraiblity
01:09:36.537 CLIENT created correlation_id=689a6fbe-1c60-47ae-826e-39332d79e99f
01:09:36.537 CLIENT SEND GET_VULNERABILITY -> s0932_vulnerability_requests
01:09:36.537 CLIENT SEND message={"correlation_id": "689a6fbe-1c60-47ae-826e-39332d79e99f", "operation": "GE
T_VULNERABILITY", "data": {"vulnerability_id": "c60fb716-d3f1-4f9e-8b24-a81ef1fb0eea"}}
01:09:36.564 CLIENT RESPONSE from s0932_vulnerability_responses (through FastAPI)
01:09:36.564 CLIENT RESPONSE message={"correlation_id": "689a6fbe-1c60-47ae-826e-39332d79e99f", "success": t
rue, "vulnerability": {"vulnerability_id": "c60fb716-d3f1-4f9e-8b24-a81ef1fb0eea", "package_name": "pip", "s
everity": "High", "vul_code": "CVE202023337", "vendor_id": 1}}

 output
retrieved vulnerability:
{
  "vulnerability_id": "c60fb716-d3f1-4f9e-8b24-a81ef1fb0eea",
  "package_name": "pip",
  "severity": "High",
  "vul_code": "CVE202023337",
  "vendor_id": 1
}
CREATE correlation_id preserved: True (69c4a536-99a3-462d-a325-0922f2f072ac)
GET correlation_id preserved: True (689a6fbe-1c60-47ae-826e-39332d79e99f)
```

3. Kafka message-flow logs: Show the message sent to <PREFIX>_entity_requests, the message received and processed by the worker, the response sent to <PREFIX>_entity_responses, and the response received by FastAPI or the client and returned to the caller.

Kafka message flow logs



4. Full sequence: Submit a log excerpt showing CREATE_ENTITY -> GET_ENTITY with the send, receive, process, reply, and final client-response lines in order.

Kafka message flow

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08> docker compose logs -
t --since 2026-11-02T09:09:36Z api worker | Sort-Object { ($_ -split '\| ',2)[1] }
s0932_api     | 2026-11-02T09:09:36.534739942Z 09:09:36.534 [API] SEND     to topic=s0932_vulnerability_requ
ests partition=0 offset=4
s0932_api     | 2026-11-02T09:09:36.534767695Z 09:09:36.534 [API] SEND     message={"correlation_id": "69c4a
536-99a3-462d-a325-0922f2f072ac", "reply_to": "s0932_vulnerability_responses", "operation": "CREATE_VULNERAB
ILITY", "data": {"package_name": "pip", "severity": "High", "vul_code": "CVE202023337", "vendor_id": 1}}
s0932_worker  | 2026-11-02T09:09:36.535485850Z 09:09:36.535 [WORKER] RECEIVE from topic=s0932_vulnerability_
requests partition=0 offset=4
s0932_worker  | 2026-11-02T09:09:36.535534464Z 09:09:36.535 [WORKER] RECEIVE message={"correlation_id": "69c
4a536-99a3-462d-a325-0922f2f072ac", "reply_to": "s0932_vulnerability_responses", "operation": "CREATE_VULNER
ABILITY", "data": {"package_name": "pip", "severity": "High", "vul_code": "CVE202023337", "vendor_id": 1}}
s0932_worker  | 2026-11-02T09:09:36.535542865Z 09:09:36.535 [WORKER] PROCESS operation=CREATE_VULNERABILITY
correlation_id=69c4a536-99a3-462d-a325-0922f2f072ac
s0932_worker  | 2026-11-02T09:09:36.536235488Z 09:09:36.536 [WORKER] REPLY    to topic=s0932_vulnerability_r
esponses partition=0 offset=4
s0932_worker  | 2026-11-02T09:09:36.536243879Z 09:09:36.536 [WORKER] REPLY    message={"correlation_id": "69
c4a536-99a3-462d-a325-0922f2f072ac", "success": true, "vulnerability_id": "c60fb716-d3f1-4f9e-8b24-a81ef1fb0
eea", "message": "Vulnerability created"}
s0932_api     | 2026-11-02T09:09:36.536815971Z 09:09:36.536 [API] RECEIVE  from topic=s0932_vulnerability_re
sponses partition=0 offset=4
s0932_api     | 2026-11-02T09:09:36.5368234922 09:09:36.536 [API] RECEIVE  message={"correlation_id": "69c4a
536-99a3-462d-a325-0922f2f072ac", "success": true, "vulnerability_id": "c60fb716-d3f1-4f9e-8b24-a81ef1fb0eea
", "message": "Vulnerability created"}
s0932_api     | 2026-11-02T09:09:36.536826802Z 09:09:36.536 [API] RETURN   response to caller for correlatio
n_id=69c4a536-99a3-462d-a325-0922f2f072ac
s0932_api     | 2026-11-02T09:09:36.53709248 6Z INFO:     172.21.0.1:43856 - "POST /vuls HTTP/1.1" 200 OK
s0932_api     | 2026-11-02T09:09:36.562455594Z 09:09:36.562 [API] SEND     to topic=s0932_vulnerability_requ
ests partition=0 offset=5
s0932_api     | 2026-11-02T09:09:36.562478296Z 09:09:36.562 [API] SEND     message={"correlation_id": "689a6
fbe-1c60-47ae-826e-39332d79e99f", "reply_to": "s0932_vulnerability_responses", "operation": "GET_VULNERABILI
TY", "data": {"vulnerability_id": "c60fb716-d3f1-4f9e-8b24-a81ef1fb0eea"}}
s0932_worker  | 2026-11-02T09:09:36.562887314Z 09:09:36.562 [WORKER] RECEIVE from topic=s0932_vulnerability_
requests partition=0 offset=5
s0932_worker  | 2026-11-02T09:09:36.562900555Z 09:09:36.562 [WORKER] RECEIVE message={"correlation_id": "689
a6fbe-1c60-47ae-826e-39332d79e99f", "reply_to": "s0932_vulnerability_responses", "operation": "GET_VULNERABI
LITY", "data": {"vulnerability_id": "c60fb716-d3f1-4f9e-8b24-a81ef1fb0eea"}}
s0932_worker  | 2026-11-02T09:09:36.562905345Z 09:09:36.562 [WORKER] PROCESS operation=GET_VULNERABILITY cor
relation_id=689a6fbe-1c60-47ae-826e-39332d79e99f
s0932_worker  | 2026-11-02T09:09:36.563867152Z 09:09:36.563 [WORKER] REPLY    to topic=s0932_vulnerability_r
esponses partition=0 offset=5
s0932_worker  | 2026-11-02T09:09:36.563877683Z 09:09:36.563 [WORKER] REPLY    message={"correlation_id": "68
9a6fbe-1c60-47ae-826e-39332d79e99f", "success": true, "vulnerability": {"vulnerability_id": "c60fb716-d3f1-4
f9e-8b24-a81ef1fb0eea", "package_name": "pip", "severity": "High", "vul_code": "CVE202023337", "vendor_id": "
1}}
s0932_api     | 2026-11-02T09:09:36.564015986Z 09:09:36.563 [API] RECEIVE  from topic=s0932_vulnerability_re
sponses partition=0 offset=5
s0932_api     | 2026-11-02T09:09:36.564025067Z 09:09:36.563 [API] RECEIVE  message={"correlation_id": "689a6
fbe-1c60-47ae-826e-39332d79e99f", "success": true, "vulnerability": {"vulnerability_id": "c60fb716-d3f1-4f9e
-8b24-a81ef1fb0eea", "package_name": "pip", "severity": "High", "vul_code": "CVE202023337", "vendor_id": 1}}
s0932_api     | 2026-11-02T09:09:36.564043658Z 09:09:36.563 [API] RETURN   response to caller for correlatio
n_id=689a6fbe-1c60-47ae-826e-39332d79e99f
s0932_api     | 2026-11-02T09:09:36.564306260Z 09:09:36.564 [API] RETURN   response to caller for correlatio
n_id=689a6fbe-1c60-47ae-826e-39332d79e99f
s0932_api     | 2026-11-02T09:09:36.564306260Z INFO:     172.21.0.1:43862 - "POST /vuls HTTP/1.1" 200 OK
```

Two topic names: s0932_vulnerability_requests and s0932_vulnerability_responses

Request and response message schemas

CREATE Schema

```
s0932_api     | 2026-11-02T09:09:36.534767695Z 09:09:36.534 [API] SEND     message={"correlation_id": "69c4a536-99a3-462d-a325-0922f2f07
2ac", "reply_to": "s0932_vulnerability_responses", "operation": "CREATE_VULNERABILITY", "data": {"package_name": "pip", "severity": "Hig
h", "vul_code": "CVE202023337", "vendor_id": 1}}
```

```
s0932_worker  | 2026-11-02T09:09:36.536243879Z 09:09:36.536 [WORKER] REPLY    message={"correlation_id": "69c4a536-99a3-462d-a325-0922f2
f072ac", "success": true, "vulnerability_id": "c60fb716-d3f1-4f9e-8b24-a81ef1fb0eea", "message": "Vulnerability created"}
```

GET Schema

```
s0932_api     | 2026-11-02T09:09:36.562478296Z 09:09:36.562 [API] SEND     message={"correlation_id": "689a6
fbe-1c60-47ae-826e-39332d79e99f", "reply_to": "s0932_vulnerability_responses", "operation": "GET_VULNERABILI
TY", "data": {"vulnerability_id": "c60fb716-d3f1-4f9e-8b24-a81ef1fb0eea"}}
```

```
s0932_worker  | 2026-11-02T09:09:36.563877683Z 09:09:36.563 [WORKER] REPLY    message={"correlation_id": "689a6fbe-1c60-47ae-826e-39332d
79e99f", "success": true, "vulnerability": {"vulnerability_id": "c60fb716-d3f1-4f9e-8b24-a81ef1fb0eea", "package_name": "pip", "severity
": "High", "vul_code": "CVE202023337", "vendor_id": 1}}
```

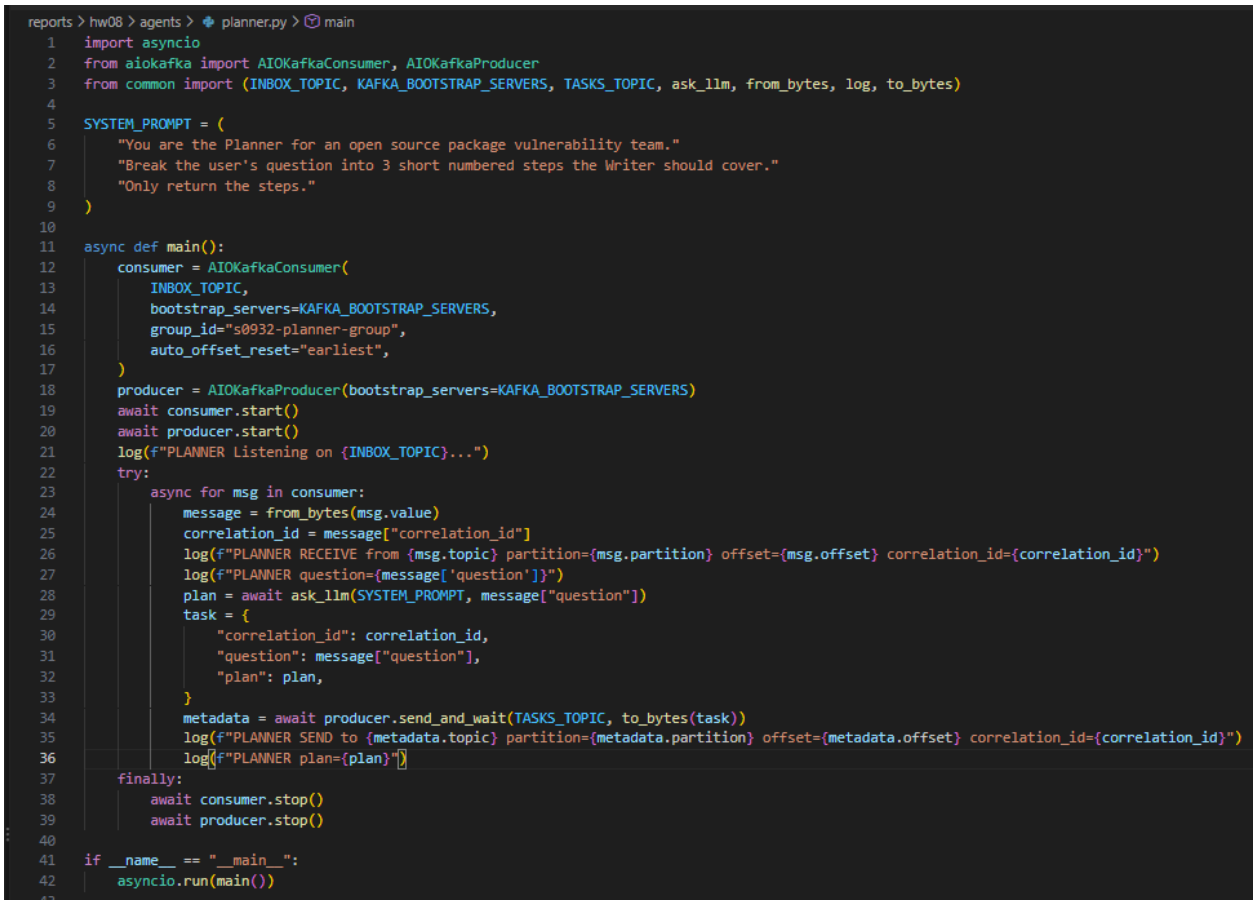
complete correlation_id flow: homework_Demo.py creates correlation_id then FastAPI puts it unchanged into the request message and then worker copies it into response and FastAPI matches the response by it and returns it and then finally the demo checks if its the same correclation_id where preserved is true.

Part 2 - Three-Agent Kafka Pipeline

Run Planner, Writer, and Reviewer as separate processes. No agent calls another agent directly -

every exchange goes through Kafka.

Planner code



```
reports > hw08 > agents > planner.py > main
import asyncio
from aiokafka import AIOKafkaConsumer, AIOKafkaProducer
from common import (INBOX_TOPIC, KAFKA_BOOTSTRAP_SERVERS, TASKS_TOPIC, ask_llm, from_bytes, log, to_bytes)

SYSTEM_PROMPT = (
    "You are the Planner for an open source package vulnerability team."
    "Break the user's question into 3 short numbered steps the Writer should cover."
    "Only return the steps."
)

async def main():
    consumer = AIOKafkaConsumer(
        INBOX_TOPIC,
        bootstrap_servers=KAFKA_BOOTSTRAP_SERVERS,
        group_id="s0932-planner-group",
        auto_offset_reset="earliest",
    )
    producer = AIOKafkaProducer(bootstrap_servers=KAFKA_BOOTSTRAP_SERVERS)
    await consumer.start()
    await producer.start()
    log(f"PLANNER Listening on {INBOX_TOPIC}...")
    try:
        async for msg in consumer:
            message = from_bytes(msg.value)
            correlation_id = message["correlation_id"]
            log(f"PLANNER RECEIVE from {msg.topic} partition={msg.partition} offset={msg.offset} correlation_id={correlation_id}")
            log(f"PLANNER question={message['question']}")
            plan = await ask_llm(SYSTEM_PROMPT, message["question"])
            task = {
                "correlation_id": correlation_id,
                "question": message["question"],
                "plan": plan,
            }
            metadata = await producer.send_and_wait(TASKS_TOPIC, to_bytes(task))
            log(f"PLANNER SEND to {metadata.topic} partition={metadata.partition} offset={metadata.offset} correlation_id={correlation_id}")
            log(f"PLANNER plan={plan}")
    finally:
        await consumer.stop()
        await producer.stop()

if __name__ == "__main__":
    asyncio.run(main())
```

Writer code

reports > hw08 > agents > writer.py > ...

```python
import asyncio
from aiokafka import AIOKafkaConsumer, AIOKafkaProducer
from common import (DRAFTS_TOPIC, KAFKA_BOOTSTRAP_SERVERS, TASKS_TOPIC, ask_llm, from_bytes, log, to_bytes)

SYSTEM_PROMPT = (
    "You are the Writer for an open source package vulnerability team."
    "Answer the question by following the plan. Keep the answer under 150 words."
)
async def main():
    consumer = AIOKafkaConsumer(
        TASKS_TOPIC,
        bootstrap_servers=KAFKA_BOOTSTRAP_SERVERS,
        group_id="s0932-writer-group",
        auto_offset_reset="earliest",
    )
    producer = AIOKafkaProducer(bootstrap_servers=KAFKA_BOOTSTRAP_SERVERS)
    await consumer.start()
    await producer.start()
    log(f"WRITER Listening on {TASKS_TOPIC}...")

    try:
        async for msg in consumer:
            task = from_bytes(msg.value)
            correlation_id = task["correlation_id"]
            log(f"WRITER RECEIVE from {msg.topic} partition={msg.partition} offset={msg.offset} correlation_id={correlation_id}")

            prompt = f"Question: {task['question']}\n\nPlan:\n{task['plan']}"
            draft = await ask_llm(SYSTEM_PROMPT, prompt)
            draft_message = {
                "correlation_id": correlation_id,
                "question": task["question"],
                "plan": task["plan"],
                "draft": draft,
            }
            metadata = await producer.send_and_wait(DRAFTS_TOPIC, to_bytes(draft_message))
            log(f"WRITER SEND to {metadata.topic} partition={metadata.partition} offset={metadata.offset} correlation_id={correlation_id}")
            log(f"WRITER draft={draft}")
    finally:
        await consumer.stop()
        await producer.stop()

if __name__ == "__main__":
    asyncio.run(main())
```

Reviewer Code

reports > hw08 > agents > reviewer.py > ...

```python
import asyncio
from aiokafka import AIOKafkaConsumer, AIOKafkaProducer
from common import (DRAFTS_TOPIC, FINAL_TOPIC, KAFKA_BOOTSTRAP_SERVERS, ask_llm, from_bytes, log, to_bytes)

SYSTEM_PROMPT = (
    "You are the Reviewer for an open source package vulnerability team. "
    "Check that the draft answers the question correctly and clearly. "
    "Reply with APPROVED or NEEDS_CHANGES on the first line, then one sentence explaining why."
)

async def main():
    consumer = AIOKafkaConsumer(
        DRAFTS_TOPIC,
        bootstrap_servers=KAFKA_BOOTSTRAP_SERVERS,
        group_id="s0932-reviewer-group",
        auto_offset_reset="earliest",
    )
    producer = AIOKafkaProducer(bootstrap_servers=KAFKA_BOOTSTRAP_SERVERS)
    await consumer.start()
    await producer.start()
    log(f"REVIEWER Listening on {DRAFTS_TOPIC}...")

    try:
        async for msg in consumer:
            draft_message = from_bytes(msg.value)
            correlation_id = draft_message["correlation_id"]
            log(f"REVIEWER RECEIVE from {msg.topic} partition={msg.partition} offset={msg.offset} correlation_id={correlation_id}")

            prompt = f"Question: {draft_message['question']}\n\nDraft:\n{draft_message['draft']}"
            review = await ask_llm(SYSTEM_PROMPT, prompt)

            #first word of the review decides the status
            if review.upper().startswith("APPROVED"):
                status = "approved"
            else:
                status = "needs_changes"

            #final JSON
            final_message = {
                "correlation_id": correlation_id,
                "status": status,
                "answer": draft_message["draft"],
                "review": review,
            }
            metadata = await producer.send_and_wait(FINAL_TOPIC, to_bytes(final_message))
            log(f"REVIEWER SEND to {metadata.topic} partition={metadata.partition} offset={metadata.offset} correlation_id={correlation_id}")
            log(f"REVIEWER status={status} review={review}")
    finally:
        await consumer.stop()
        await producer.stop()

if __name__ == "__main__":
    asyncio.run(main())
```

I. Topics and Workflow

One complete message journey from <PREFIX> _inbox to <PREFIX> _final, shown as log lines for each stage.

```
02:30:08.871 YOU SEND to s0932_inbox partition=0 offset=2 correlation_id=aa4b70a6-0ee1-409e-a2
25-9bdfab829b0a
```

```
02:29:40.164 PLANNER RECEIVE from s0932_inbox partition=0 offset=1 correlation_id=e63b6038-0a
af-443e-83ea-ed0c2e7f4d7d
```

```
02:29:43.578 PLANNER SEND to s0932_tasks partition=1 offset=0 correlation_id=e63b6038-0aaf-44
3e-83ea-ed0c2e7f4d7d
```

```
02:29:54.385 WRITER RECEIVE from s0932_tasks partition=1 offset=0 correlation_id=e63b6038-0aaf
-443e-83ea-ed0c2e7f4d7d
```

```
02:29:55.969 WRITER SEND to s0932_drafts partition=0 offset=1 correlation_id=e63b6038-0aaf-44
e-83ea-ed0c2e7f4d7d
```

```
02:30:04.543 REVIEWER RECEIVE from s0932_drafts partition=0 offset=1 correlation_id=e63b6038-
0aaf-443e-83ea-ed0c2e7f4d7d
02:30:05.420 REVIEWER SEND to s0932_final partition=0 offset=1 correlation_id=e63b6038-0aaf-4
43e-83ea-ed0c2e7f4d7d
```

```
02:30:11.276 YOU RECEIVE from s0932_final partition=0 offset=2 correlation_id=aa4b70a6-0ee1-40
9e-a225-9bdfab829b0a

Final JSON message:
{
  "correlation_id": "aa4b70a6-0ee1-409e-a225-9bdfab829b0a",
  "status": "approved",
  "answer": "Our team should first verify if CVE-2020-23337 affects the pip version currently
in use. If confirmed, assess the vulnerability's impact on the project and its dependencies. N
ext, plan an immediate update to a patched pip version, ensuring compatibility with existing p
ackages. Finally, monitor the system for any issues post-update and consider implementing autc
mated checks to prevent future vulnerabilities.",
  "review": "APPROVED  \nThe draft provides a clear, actionable plan for addressing the vulner
ability, covering verification, impact assessment, remediation, and ongoing monitoring."
}
```

II. Kafka Topic Creation

Use your Kafka CLI or UI to create the four topics.

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08> docker exec kafka /opt/kafka/bin/kafka-topics.sh --create --if-not-exists --topic s0932_inbox --partition
s 1 --replication-factor 1 --bootstrap-server localhost:9092
WARNING: Due to limitations in metric names, topics with a period ('.') or underscore ('_') could collide. To avoid issues it is best to use either, but not both.
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08> docker exec kafka /opt/kafka/bin/kafka-topics.sh --create --if-not-exists --topic s0932_tasks --partition
s 2 --replication-factor 1 --bootstrap-server localhost:9092
WARNING: Due to limitations in metric names, topics with a period ('.') or underscore ('_') could collide. To avoid issues it is best to use either, but not both.
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08> docker exec kafka /opt/kafka/bin/kafka-topics.sh --create --if-not-exists --topic s0932_drafts --partitio
ns 1 --replication-factor 1 --bootstrap-server localhost:9092
WARNING: Due to limitations in metric names, topics with a period ('.') or underscore ('_') could collide. To avoid issues it is best to use either, but not both.
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08> docker exec kafka /opt/kafka/bin/kafka-topics.sh --create --if-not-exists --topic s0932_final --partition
s 1 --replication-factor 1 --bootstrap-server localhost:9092
WARNING: Due to limitations in metric names, topics with a period ('.') or underscore ('_') could collide. To avoid issues it is best to use either, but not both.
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08> docker exec kafka /opt/kafka/bin/kafka-topics.sh --list --bootstrap-server localhost:9092
__consumer_offsets
s0932_drafts
s0932_final
s0932_inbox
s0932_tasks
s0932_vulnerability_requests
s0932_vulnerability_responses
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08> docker exec kafka /opt/kafka/bin/kafka-topics.sh --describe --bootstrap-server localhost:9092 --topic s09
32_inbox,s0932_tasks,s0932_drafts,s0932_final
Topic: s0932_drafts     TopicId: YNq7jbRBTNagp7hh_B5NDQ PartitionCount: 1       ReplicationFactor: 1    Configs:
        Topic: s0932_drafts     Partition: 0    Leader: 1       Replicas: 1     Isr: 1  Elr:    LastKnownElr:
Topic: s0932_inbox      TopicId: TKyc1-98QviiGY2U3OyRyQ PartitionCount: 1       ReplicationFactor: 1    Configs:
        Topic: s0932_inbox      Partition: 0    Leader: 1       Replicas: 1     Isr: 1  Elr:    LastKnownElr:
Topic: s0932_tasks      TopicId: 1Hg89dsSQXeeYgDJ0uWANw PartitionCount: 2       ReplicationFactor: 1    Configs:
        Topic: s0932_tasks      Partition: 0    Leader: 1       Replicas: 1     Isr: 1  Elr:    LastKnownElr:
        Topic: s0932_tasks      Partition: 1    Leader: 1       Replicas: 1     Isr: 1  Elr:    LastKnownElr:
Topic: s0932_final      TopicId: 18H1RRtOSU-NQDfUEIq0gw PartitionCount: 1       ReplicationFactor: 1    Configs:
        Topic: s0932_final      Partition: 0    Leader: 1       Replicas: 1     Isr: 1  Elr:    LastKnownElr:
```

III. Running the Agents

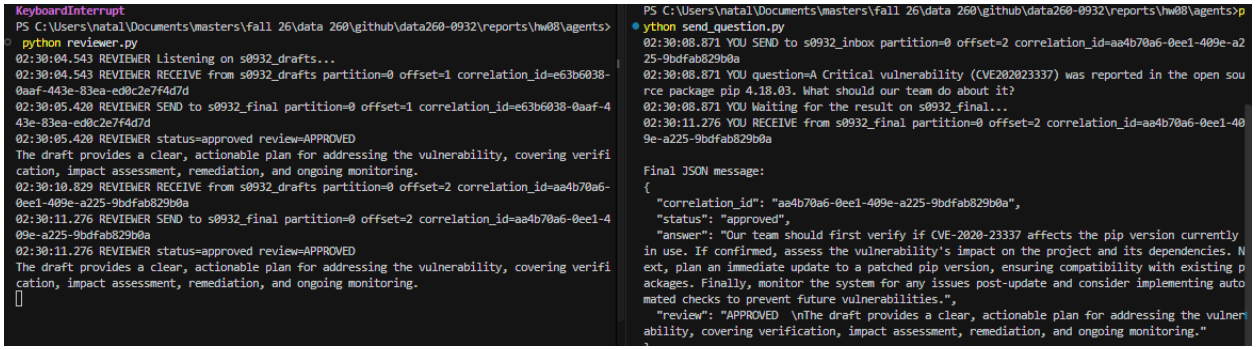
Run each agent in its own terminal:

Terminal 1 python planner.py/Terminal 2 python writer.py

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents>
 python planner.py
02:29:40.164 PLANNER Listening on s0932_inbox...
02:29:40.164 PLANNER RECEIVE from s0932_inbox partition=0 offset=1 correlation_id=e63b6038-0a
af-443e-83ea-ed0c2e7f4d7d
02:29:40.164 PLANNER question=A Critical vulnerability (CVE202023337) was reported in the ope
n source package pip 4.18.03. What should our team do about it?
02:29:43.578 PLANNER SEND to s0932_tasks partition=1 offset=0 correlation_id=e63b6038-0aaf-44
3e-83ea-ed0c2e7f4d7d
02:29:43.578 PLANNER plan=1. Verify the vulnerability details and confirm if it affects the v
ersion of pip in use.
2. Assess the impact of the vulnerability on the current project and dependencies.
3. Plan and execute an update to a patched version of pip, and monitor for any related issues
```

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents>
 python writer.py
02:29:54.384 WRITER Listening on s0932_tasks...
02:29:54.385 WRITER RECEIVE from s0932_tasks partition=1 offset=0 correlation_id=e63b6038-0aa
f-443e-83ea-ed0c2e7f4d7d
02:29:55.969 WRITER SEND to s0932_drafts partition=0 offset=1 correlation_id=e63b6038-0aaf-44
3e-83ea-ed0c2e7f4d7d
02:29:55.969 WRITER draft=Our team should first verify if CVE-2020-23337 affects the pip vers
ion currently in use. If confirmed, assess the vulnerability's impact on the project and its
dependencies. Next, plan an immediate update to a patched pip version, ensuring compatibility
 with existing packages. Finally, monitor the system for any issues post-update and consider
implementing automated checks to prevent future vulnerabilities.
```

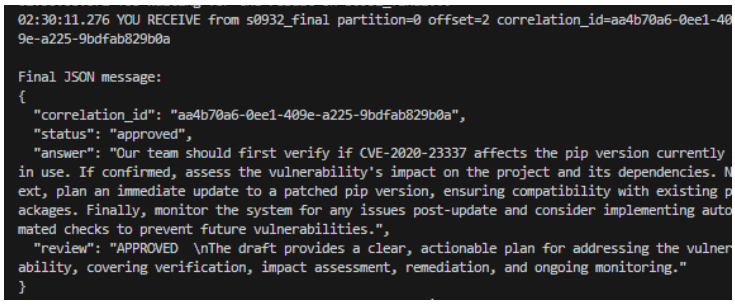
Terminal 3 python reviewer.py/Terminal 4 python send_question.py



IV. Result

The final message is JSON with status and answer. Preserve correlation_id through every stage. Read the result from <PREFIX>_final.

The final JSON message with its status and answer



A short report explaining how Kafka helped your agents coordinate.

Kafka let my planner, writer, and reviewer work together without having to call each other directly. Each agent ran its own program in its own terminal and only talked to Kafka topics. The planner read questions from s0932_inbox and sent a plan to s0932_tasks. Afterwards, the writer read that plan and sent a draft to s0932_drafts. Finally the reviewer read the draft and sent the final answer to s0932_final. Because each agent only knows its own input and output topics, I am able to start, stop, or change one agent without it affecting another. Kafka also stores the messages, so these messages wait in the topic until the agent starts. In my run, the writer started almost two minutes after the planner. Every message also carried the same correlation_id, with the value of 260389d2-5c26-4dc5-b67c-879c7eec7bd3. It was the same for the four separate messages so in the end running the send_question.py got the same correlation_id result in s0932 final.

Part 3 - Failure Behavior

I. Duplicate Delivery

Replay a message with the same correlation_id, then add idempotency so replay causes no extra side effect.

Duplicate before fix terminal output

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932> cd .\reports\hw08
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08> cd .\agents\
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents> python .\writer.py --before
11:48:53.705 [WRITER-A] Listening on s0932_tasks... mode=BEFORE FIX (no idempotency, auto commit, no retries)
11:49:11.167 [WRITER-A] RECEIVE from s0932_tasks partition=0 offset=2
11:49:11.167 [WRITER-A] correlation_id=4dd7ca80-9e96-4cc2-a386-c7096a342591
11:49:18.302 [WRITER-A] SEND to s0932_drafts partition=0 offset=3 correlation_id=4dd7ca80-9e96-4cc2-a386-c7096a342591
11:49:18.302 [WRITER-A] draft=CVE-2018-18074 allows remote code execution via crafted HTTP headers, posing a serious security risk. The vulnerability is fixed in **requests 2.21.0** and later. You should upgrade **immediately** to a version >= 2.21.0 to mitigate this risk. Delaying the upgrade could expose your system to potential attacks. Ensure all dependent packages are also updated to avoid compatibility issues.
11:49:18.801 [WRITER-A] RECEIVE from s0932_tasks partition=0 offset=3
11:49:18.801 [WRITER-A] correlation_id=4dd7ca80-9e96-4cc2-a386-c7096a342591
11:49:19.897 [WRITER-A] SEND to s0932_drafts partition=0 offset=4 correlation_id=4dd7ca80-9e96-4cc2-a386-c7096a342591
11:49:19.897 [WRITER-A] draft=CVE-2018-18074 allows remote code execution via crafted HTTP headers, posing a serious security risk. The vulnerability is fixed in **requests 2.20.0**. You should upgrade **immediately** to mitigate this risk and protect your application from potential attacks. Delaying the upgrade could expose your system to exploitation. Ensure all dependencies are compatible with the new version before updating.
```

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08> cd .\agents\
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents>python .\send_test_task.py duplicate
11:49:11.167 [TEST] SEND task correlation_id=4dd7ca80-9e96-4cc2-a386-c7096a342591 (first time) to s0932_tasks partition=0 offset=2
11:49:11.168 [TEST] SEND task correlation_id=4dd7ca80-9e96-4cc2-a386-c7096a342591 (REPLAY) to s0932_tasks partition=0 offset=3
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents>
```

Duplicate after fix terminal output

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents> python .\writer.py
11:51:00.415 [WRITER-A] Listening on s0932_tasks... mode=AFTER FIX (idempotency, commit after success, 3 retries + dead letter)
11:51:05.188 [WRITER-A] RECEIVE from s0932_tasks partition=0 offset=4
11:51:05.188 [WRITER-A] correlation_id=40d444c4-47cb-49f5-a6fc-0b06bc0174ee
11:51:06.688 [WRITER-A] SEND to s0932_drafts partition=0 offset=5 correlation_id=40d444c4-47cb-49f5-a6fc-0b06bc0174ee
11:51:06.688 [WRITER-A] draft=CVE-2018-18074 allows remote code execution via crafted HTTP headers, posing a serious security risk. The vulnerability is fixed in **requests 2.20.0**. You should upgrade **immediately** to mitigate this risk and protect your application from potential attacks. Delaying the upgrade could expose your system to exploitation. Ensure all dependencies are compatible with the new version before updating.
11:51:06.689 [WRITER-A] COMMIT partition=0 offset=4 (next offset to read: 5)
11:51:06.795 [WRITER-A] RECEIVE from s0932_tasks partition=0 offset=5
11:51:06.795 [WRITER-A] correlation_id=40d444c4-47cb-49f5-a6fc-0b06bc0174ee
11:51:06.795 [WRITER-A] DUPLICATE correlation_id=40d444c4-47cb-49f5-a6fc-0b06bc0174ee was already processed -> skipped, no new draft
11:51:06.796 [WRITER-A] COMMIT partition=0 offset=5 (next offset to read: 6)
```

```
ents\
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents>python .\send_test_task.py duplicate
11:49:11.167 [TEST] SEND task correlation_id=4dd7ca80-9e96-4cc2-a386-c7096a342591 (first time) to s0932_tasks partition=0 offset=2
11:49:11.168 [TEST] SEND task correlation_id=4dd7ca80-9e96-4cc2-a386-c7096a342591 (REPLAY) to s0932_tasks partition=0 offset=3
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents>python .\send_test_task.py duplicate
11:51:05.188 TEST SEND task correlation_id=40d444c4-47cb-49f5-a6fc-0b06bc0174ee (first time) to s0932_tasks partition=0 offset=4
11:51:05.189 TEST SEND task correlation_id=40d444c4-47cb-49f5-a6fc-0b06bc0174ee (REPLAY) to s0932_tasks partition=0 offset=5
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents>
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents>
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents>
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents>
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents>
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents>
```

II. Consumer Death

Terminate Writer mid-processing and record what happens. After restarting Writer, show that the message is redelivered or completed, and ensure the offset is committed only after successful processing.

Consumer death before fix

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents> python .\writer.py --before --slow
11:56:35.906 [WRITER-A] Listening on s0932_tasks... mode=BEFORE FIX (no idempotency, auto commit, no retries)
11:56:35.906 [WRITER-A] RECEIVE from s0932_tasks partition=0 offset=7
11:56:35.906 [WRITER-A] correlation_id=79c4724c-1246-4b2c-84a6-004592ea4090
11:56:35.906 [WRITER-A] PROCESSING... waiting 15 seconds (press Ctrl+C now to kill the writer mid-processing)
Traceback (most recent call last):
  File "C:\Python314\Lib\asyncio\runners.py", line 128, in run
    return self._loop.run_until_complete(task)
           ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^^^^
  File "C:\Python314\Lib\asyncio\base_events.py", line 720, in run_until_complete
    return future.result()
           ~~~~~~~~~~~~~^^
  File "C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents\writer.py", line 118, in main
    await handle_task(msg, producer, processed_ids)
  File "C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents\writer.py", line 79, in handle_task
    await asyncio.sleep(15)
  File "C:\Python314\Lib\asyncio\tasks.py", line 705, in sleep
    return await future
           ^^^^^^^^^^^^
asyncio.exceptions.CancelledError

During handling of the above exception, another exception occurred:

Traceback (most recent call last):
  File "C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents\writer.py", line 141, in <module>
    asyncio.run(main())
    ~~~~~~~~~~~^^^^^^^^
  File "C:\Python314\Lib\asyncio\runners.py", line 205, in run
    return runner.run(main)
           ~~~~~~~~~~^^^^^^
  File "C:\Python314\Lib\asyncio\runners.py", line 133, in run
    raise KeyboardInterrupt()
KeyboardInterrupt
```

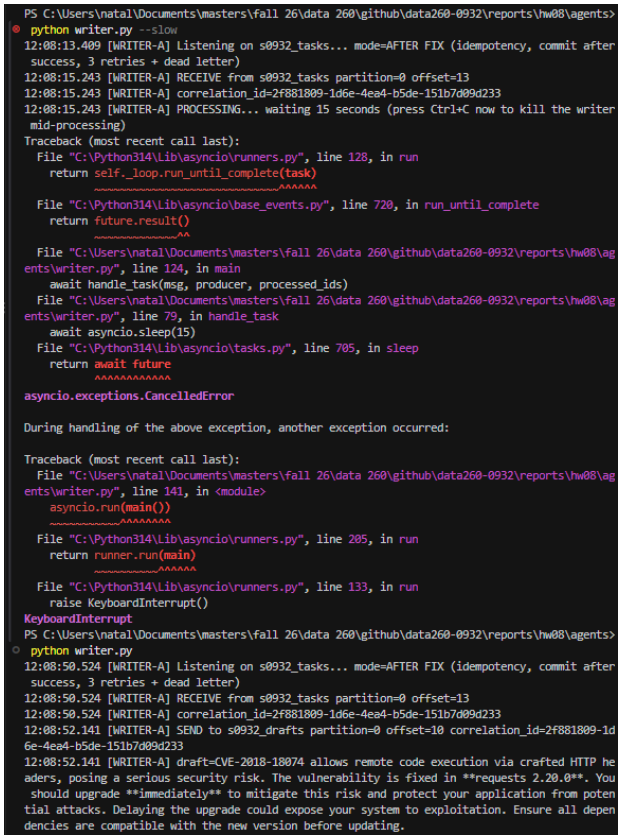
```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents>docker exec kafka /opt/kafka/bin/kafka-consumer-groups.sh --bootstrap-server localhost:9092 --describe --group s0932-writer-group

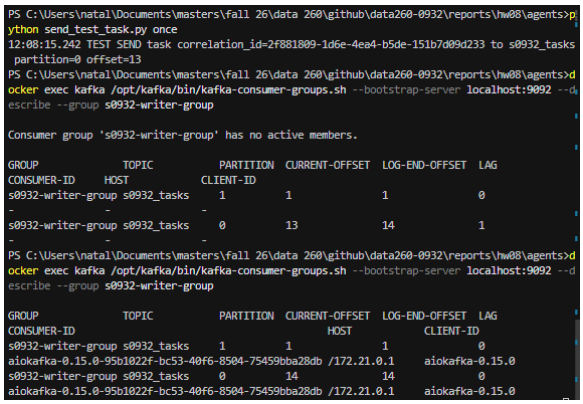
Consumer group 's0932-writer-group' has no active members.

GROUP              TOPIC           PARTITION  CURRENT-OFFSET  LOG-END-OFFSET  LAG             CONSUMER-ID     HOST            CLIENT-ID
s0932-writer-group s0932_tasks     1          1               1               0               -               -               -
s0932-writer-group s0932_tasks     0          8               8               0               -               -               -
```

```
python .\writer.py --before
11:58:17.458 [WRITER-A] Listening on s0932_tasks... mode=BEFORE FIX (no idempotency, auto commit, no retries)
```

Consumer death after fix





III. Poison Message

Add bounded retries and a PREFIX-prefixed dead-letter topic named <PREFIX>_dead_letter. Show that a bad message is retried up to the limit, moved to the dead-letter topic, and no longer blocks later messages.

Before poison message fix

```
python .\writer.py --before
12:03:23.833 [WRITER-A] Listening on s0932_tasks... mode=BEFORE FIX (no idempotency, auto commit, no retries)
12:03:23.833 [WRITER-A] RECEIVE from s0932_tasks partition=0 offset=8
12:03:23.833 [WRITER-A] correlation_id=b21119f4-17b8-459f-9d78-db6e90df4ea7
12:03:27.313 [WRITER-A] SEND to s0932_drafts partition=0 offset=7 correlation_id=b21119f4-17b8-459f-9d78-db6e90df4ea7
12:03:27.313 [WRITER-A] draft=CVE-2018-18074 allows remote code execution via crafted HTTP headers, posing a serious security risk. The vulnerability is fixed in **requests 2.21.0** and later. You should upgrade **immediately** to a version >= 2.21.0 to mitigate this risk. Delaying the upgrade could expose your system to potential attacks. Ensure all dependent packages are also updated to avoid compatibility issues.
12:03:27.313 [WRITER-A] RECEIVE from s0932_tasks partition=0 offset=9
Traceback (most recent call last):
  File "C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents\writer.py", line 141, in <module>
    asyncio.run(main())
    ~~~~~~~~~~~^^^^^^^^
  File "C:\Python314\Lib\asyncio\runners.py", line 205, in run
    return runner.run(main)
           ~~~~~~~~~~^^^^^^
  File "C:\Python314\Lib\asyncio\runners.py", line 128, in run
    return self._loop.run_until_complete(task)
           ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^^^^
  File "C:\Python314\Lib\asyncio\base_events.py", line 720, in run_until_complete
    return future.result()
           ~~~~~~~~~~~~~^^
  File "C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents\writer.py", line 118, in main
    await handle_task(msg, producer, processed_ids)
  File "C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents\writer.py", line 62, in handle_task
    task = from_bytes(msg.value) # if not JSON then fail here so checking
  File "C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents\common.py", line 24, in from_bytes
    return json.loads(raw.decode())
           ~~~~~~~~~~^^^^^^^^^^^^^
  File "C:\Python314\Lib\json\__init__.py", line 352, in loads
    return _default_decoder.decode(s)
           ~~~~~~~~~~~~~~~~~~~~~~~^^^
  File "C:\Python314\Lib\json\decoder.py", line 345, in decode
    obj, end = self.raw_decode(s, idx=_w(s, 0).end())
               ~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Python314\Lib\json\decoder.py", line 363, in raw_decode
    raise JSONDecodeError("Expecting value", s, err.value) from None
```

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents>p
ython .\send_test_task.py poison
12:03:17.890 TEST SEND POISON message to s0932_tasks partition=0 offset=9
12:03:17.891 TEST SEND normal task after the poison, correlation_id=4604a542-c380-4798-b149-4a288f30a75b to s0932_tasks partition=0 offset=10
```

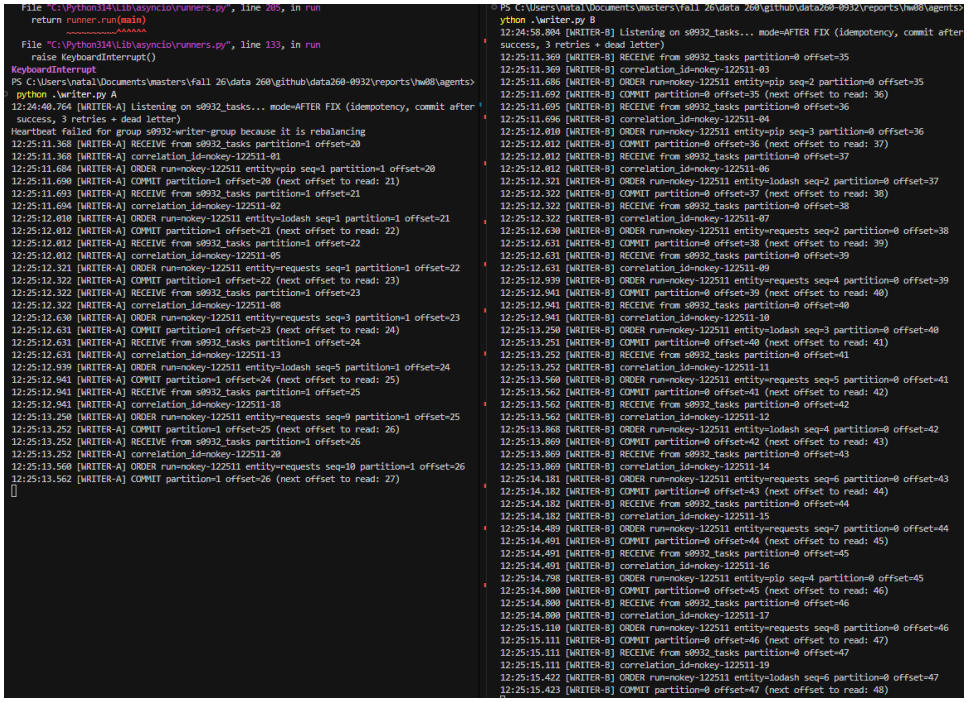
After poison message fix

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents>
 python .\writer.py
12:04:38.833 [WRITER-A] Listening on s0932_tasks... mode=AFTER FIX (idempotency, commit after success, 3 retries + dead letter)
12:04:38.833 [WRITER-A] RECEIVE from s0932_tasks partition=0 offset=10
12:04:38.833 [WRITER-A] correlation_id=4604a542-c380-4798-b149-4a288f30a75b
12:04:40.107 [WRITER-A] SEND to s0932_drafts partition=0 offset=8 correlation_id=4604a542-c380-4798-b149-4a288f30a75b
12:04:40.107 [WRITER-A] draft=CVE-2018-18074 allows remote code execution via crafted HTTP headers, posing a serious security risk. The vulnerability is fixed in **requests 2.20.0**. You should upgrade **immediately** to mitigate this risk and protect your application from potential attacks. Delaying the upgrade could expose your system to exploitation. Ensure all dependencies are compatible with the new version before updating.
12:04:40.108 [WRITER-A] COMMIT partition=0 offset=10 (next offset to read: 11)
12:04:49.963 [WRITER-A] RECEIVE from s0932_tasks partition=0 offset=11
12:04:49.964 [WRITER-A] RETRY attempt 1/3 failed: JSONDecodeError('Expecting value: line 1 column 1 (char 0)')
12:04:50.967 [WRITER-A] RETRY attempt 2/3 failed: JSONDecodeError('Expecting value: line 1 column 1 (char 0)')
12:04:51.968 [WRITER-A] RETRY attempt 3/3 failed: JSONDecodeError('Expecting value: line 1 column 1 (char 0)')
12:04:51.973 [WRITER-A] DEAD LETTER sent to s0932_dead_letter partition=0 offset=0
12:04:51.973 [WRITER-A] DEAD LETTER message={'source_topic': 's0932_tasks', 'partition': 0, 'offset': 11, 'original_message': 'this is not valid JSON {', 'error': "JSONDecodeError('Expecting value: line 1 column 1 (char 0)')", 'attempts': 3}
12:04:51.974 [WRITER-A] COMMIT partition=0 offset=11 (next offset to read: 12)
12:04:52.082 [WRITER-A] RECEIVE from s0932_tasks partition=0 offset=12
12:04:52.082 [WRITER-A] correlation_id=0f168683-828d-4c1a-96bd-4cd097442168
12:04:53.635 [WRITER-A] SEND to s0932_drafts partition=0 offset=9 correlation_id=0f168683-828d-4c1a-96bd-4cd097442168
12:04:53.636 [WRITER-A] draft=CVE-2018-18074 allows remote code execution via crafted HTTP headers, posing a serious security risk. The vulnerability is fixed in **requests 2.20.0**. You should upgrade **immediately** to mitigate this risk and protect your application from potential attacks. Delaying the upgrade could expose your system to exploitation. Ensure all dependencies are compatible with the new version before updating.
12:04:53.637 [WRITER-A] COMMIT partition=0 offset=12 (next offset to read: 13)
```

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents>p
ython .\send_test_task.py poison
12:04:49.963 TEST SEND POISON message to s0932_tasks partition=0 offset=11
12:04:49.964 TEST SEND normal task after the poison, correlation_id=0f168683-828d-4c1a-96bd-4cd097442168 to s0932_tasks partition=0 offset=12
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents>
 docker exec kafka /opt/kafka/bin/kafka-console-consumer.sh --bootstrap-server localhost:9092
--topic s0932_dead_letter --from-beginning --timeout-ms 5000
{"source_topic": "s0932_tasks", "partition": 0, "offset": 11, "original_message": "this is not valid JSON {", "error": "JSONDecodeError('Expecting value: line 1 column 1 (char 0)')", "attempts": 3}
[2026-11-02 20:05:27,984] ERROR Error processing message, terminating consumer process:  (org.apache.kafka.tools.consumer.ConsoleConsumer)
org.apache.kafka.common.errors.TimeoutException
Processed a total of 1 messages
```

IV. Ordering

Ordering before fix



```
  File "C:\Python314\Lib\asyncio\runners.py", line 204, in run
    return runner.run(main)
           ~~~~~~~~~~^^^^^^
  File "C:\Python314\Lib\asyncio\runners.py", line 133, in run
    raise KeyboardInterrupt()
KeyboardInterrupt
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents> python .\writer.py A
12:24:40.764 [WRITER-A] Listening on s0932_tasks... mode=AFTER FIX (idempotency, commit after success, 3 retries + dead letter)
Heartbeat failed for group s0932-writer-group because it is rebalancing
12:25:11.368 [WRITER-A] RECEIVE from s0932_tasks partition=1 offset=20
12:25:11.368 [WRITER-A] correlation_id=nokey-122511-01
12:25:11.684 [WRITER-A] ORDER run=nokey-122511 entity=pip seq=1 partition=1 offset=20
12:25:11.690 [WRITER-A] COMMIT partition=1 offset=20 (next offset to read: 21)
12:25:11.693 [WRITER-A] RECEIVE from s0932_tasks partition=1 offset=21
12:25:11.694 [WRITER-A] correlation_id=nokey-122511-02
12:25:12.010 [WRITER-A] ORDER run=nokey-122511 entity=lodash seq=1 partition=1 offset=21
12:25:12.012 [WRITER-A] COMMIT partition=1 offset=21 (next offset to read: 22)
12:25:12.012 [WRITER-A] RECEIVE from s0932_tasks partition=1 offset=22
12:25:12.012 [WRITER-A] correlation_id=nokey-122511-05
12:25:12.321 [WRITER-A] ORDER run=nokey-122511 entity=requests seq=1 partition=1 offset=22
12:25:12.322 [WRITER-A] COMMIT partition=1 offset=22 (next offset to read: 23)
12:25:12.322 [WRITER-A] RECEIVE from s0932_tasks partition=1 offset=23
12:25:12.322 [WRITER-A] correlation_id=nokey-122511-08
12:25:12.630 [WRITER-A] ORDER run=nokey-122511 entity=requests seq=3 partition=1 offset=23
12:25:12.631 [WRITER-A] COMMIT partition=1 offset=23 (next offset to read: 24)
12:25:12.631 [WRITER-A] RECEIVE from s0932_tasks partition=1 offset=24
12:25:12.631 [WRITER-A] correlation_id=nokey-122511-13
12:25:12.939 [WRITER-A] ORDER run=nokey-122511 entity=lodash seq=5 partition=1 offset=24
12:25:12.941 [WRITER-A] COMMIT partition=1 offset=24 (next offset to read: 25)
12:25:12.941 [WRITER-A] RECEIVE from s0932_tasks partition=1 offset=25
12:25:12.941 [WRITER-A] correlation_id=nokey-122511-18
12:25:13.250 [WRITER-A] ORDER run=nokey-122511 entity=requests seq=9 partition=1 offset=25
12:25:13.252 [WRITER-A] COMMIT partition=1 offset=25 (next offset to read: 26)
12:25:13.252 [WRITER-A] RECEIVE from s0932_tasks partition=1 offset=26
12:25:13.252 [WRITER-A] correlation_id=nokey-122511-20
12:25:13.560 [WRITER-A] ORDER run=nokey-122511 entity=requests seq=10 partition=1 offset=26
12:25:13.562 [WRITER-A] COMMIT partition=1 offset=26 (next offset to read: 27)
```

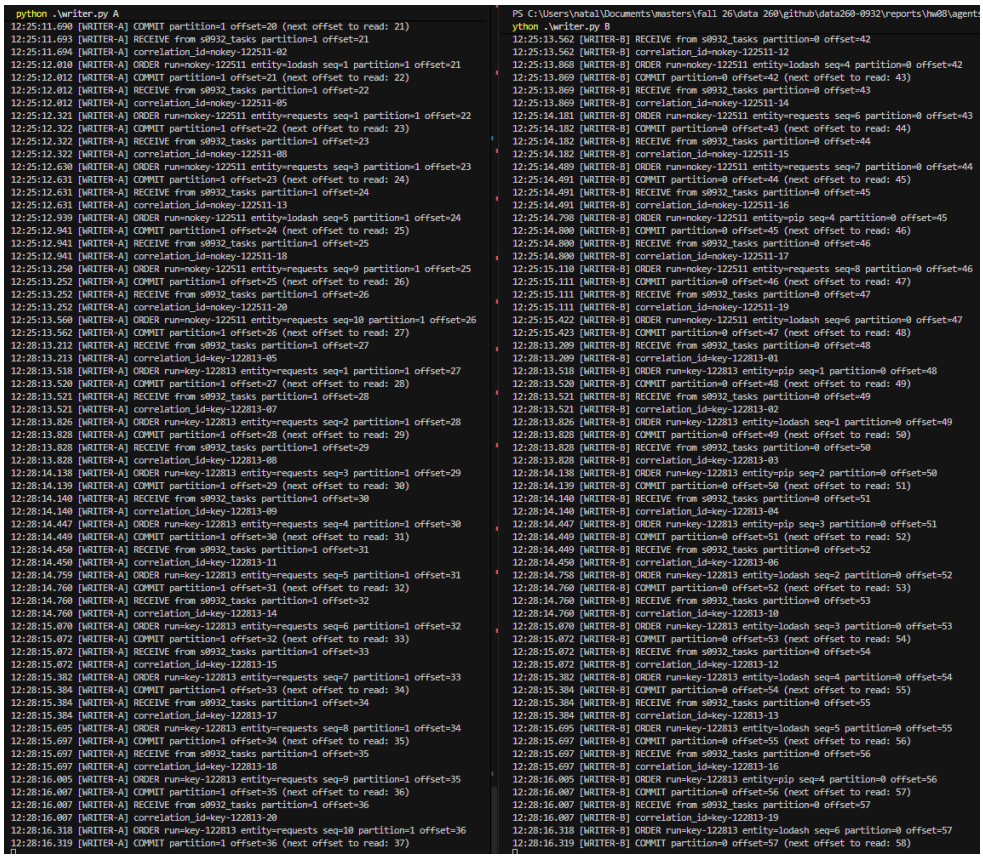
```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents>
python .\writer.py B
12:24:58.804 [WRITER-B] Listening on s0932_tasks... mode=AFTER FIX (idempotency, commit after success, 3 retries + dead letter)
12:25:11.369 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=35
12:25:11.369 [WRITER-B] correlation_id=nokey-122511-03
12:25:11.686 [WRITER-B] ORDER run=nokey-122511 entity=pip seq=2 partition=0 offset=35
12:25:11.692 [WRITER-B] COMMIT partition=0 offset=35 (next offset to read: 36)
12:25:11.695 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=36
12:25:11.696 [WRITER-B] correlation_id=nokey-122511-04
12:25:12.010 [WRITER-B] ORDER run=nokey-122511 entity=pip seq=3 partition=0 offset=36
12:25:12.012 [WRITER-B] COMMIT partition=0 offset=36 (next offset to read: 37)
12:25:12.012 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=37
12:25:12.012 [WRITER-B] correlation_id=nokey-122511-06
12:25:12.321 [WRITER-B] ORDER run=nokey-122511 entity=lodash seq=2 partition=0 offset=37
12:25:12.322 [WRITER-B] COMMIT partition=0 offset=37 (next offset to read: 38)
12:25:12.322 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=38
12:25:12.322 [WRITER-B] correlation_id=nokey-122511-07
12:25:12.630 [WRITER-B] ORDER run=nokey-122511 entity=requests seq=2 partition=0 offset=38
12:25:12.631 [WRITER-B] COMMIT partition=0 offset=38 (next offset to read: 39)
12:25:12.631 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=39
12:25:12.631 [WRITER-B] correlation_id=nokey-122511-09
12:25:12.939 [WRITER-B] ORDER run=nokey-122511 entity=requests seq=4 partition=0 offset=39
12:25:12.941 [WRITER-B] COMMIT partition=0 offset=39 (next offset to read: 40)
12:25:12.941 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=40
12:25:12.941 [WRITER-B] correlation_id=nokey-122511-10
12:25:13.250 [WRITER-B] ORDER run=nokey-122511 entity=lodash seq=3 partition=0 offset=40
12:25:13.251 [WRITER-B] COMMIT partition=0 offset=40 (next offset to read: 41)
12:25:13.252 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=41
12:25:13.252 [WRITER-B] correlation_id=nokey-122511-11
12:25:13.560 [WRITER-B] ORDER run=nokey-122511 entity=requests seq=5 partition=0 offset=41
12:25:13.562 [WRITER-B] COMMIT partition=0 offset=41 (next offset to read: 42)
12:25:13.562 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=42
12:25:13.562 [WRITER-B] correlation_id=nokey-122511-12
12:25:13.868 [WRITER-B] ORDER run=nokey-122511 entity=lodash seq=4 partition=0 offset=42
12:25:13.869 [WRITER-B] COMMIT partition=0 offset=42 (next offset to read: 43)
12:25:13.869 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=43
12:25:13.869 [WRITER-B] correlation_id=nokey-122511-14
12:25:14.181 [WRITER-B] ORDER run=nokey-122511 entity=requests seq=6 partition=0 offset=43
12:25:14.182 [WRITER-B] COMMIT partition=0 offset=43 (next offset to read: 44)
12:25:14.182 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=44
12:25:14.182 [WRITER-B] correlation_id=nokey-122511-15
12:25:14.489 [WRITER-B] ORDER run=nokey-122511 entity=requests seq=7 partition=0 offset=44
12:25:14.491 [WRITER-B] COMMIT partition=0 offset=44 (next offset to read: 45)
12:25:14.491 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=45
12:25:14.491 [WRITER-B] correlation_id=nokey-122511-16
12:25:14.798 [WRITER-B] ORDER run=nokey-122511 entity=pip seq=4 partition=0 offset=45
12:25:14.800 [WRITER-B] COMMIT partition=0 offset=45 (next offset to read: 46)
12:25:14.800 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=46
12:25:14.800 [WRITER-B] correlation_id=nokey-122511-17
12:25:15.110 [WRITER-B] ORDER run=nokey-122511 entity=requests seq=8 partition=0 offset=46
12:25:15.111 [WRITER-B] COMMIT partition=0 offset=46 (next offset to read: 47)
12:25:15.111 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=47
12:25:15.111 [WRITER-B] correlation_id=nokey-122511-19
12:25:15.422 [WRITER-B] ORDER run=nokey-122511 entity=lodash seq=6 partition=0 offset=47
12:25:15.423 [WRITER-B] COMMIT partition=0 offset=47 (next offset to read: 48)
```

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents> python .\ordering_test.py nokey
12:25:11.352 [TEST] run=nokey-122511 VERIFY_SEED=260932 messages=20
12:25:11.367 [TEST] SEND nokey-122511-01 entity=pip seq=1 key=None -> partition=1
12:25:11.368 [TEST] SEND nokey-122511-02 entity=lodash seq=1 key=None -> partition=1
12:25:11.369 [TEST] SEND nokey-122511-03 entity=pip seq=2 key=None -> partition=0
12:25:11.370 [TEST] SEND nokey-122511-04 entity=pip seq=3 key=None -> partition=0
12:25:11.371 [TEST] SEND nokey-122511-05 entity=requests seq=1 key=None -> partition=1
12:25:11.371 [TEST] SEND nokey-122511-06 entity=lodash seq=2 key=None -> partition=0
12:25:11.372 [TEST] SEND nokey-122511-07 entity=requests seq=2 key=None -> partition=0
12:25:11.373 [TEST] SEND nokey-122511-08 entity=requests seq=3 key=None -> partition=1
12:25:11.374 [TEST] SEND nokey-122511-09 entity=requests seq=4 key=None -> partition=0
12:25:11.375 [TEST] SEND nokey-122511-10 entity=lodash seq=3 key=None -> partition=0
12:25:11.375 [TEST] SEND nokey-122511-11 entity=requests seq=5 key=None -> partition=0
12:25:11.376 [TEST] SEND nokey-122511-12 entity=lodash seq=4 key=None -> partition=0
12:25:11.377 [TEST] SEND nokey-122511-13 entity=lodash seq=5 key=None -> partition=1
12:25:11.378 [TEST] SEND nokey-122511-14 entity=requests seq=6 key=None -> partition=0
12:25:11.379 [TEST] SEND nokey-122511-15 entity=requests seq=7 key=None -> partition=0
12:25:11.379 [TEST] SEND nokey-122511-16 entity=pip seq=4 key=None -> partition=0
12:25:11.380 [TEST] SEND nokey-122511-17 entity=requests seq=8 key=None -> partition=0
12:25:11.381 [TEST] SEND nokey-122511-18 entity=requests seq=9 key=None -> partition=1
12:25:11.382 [TEST] SEND nokey-122511-19 entity=lodash seq=6 key=None -> partition=0
12:25:11.383 [TEST] SEND nokey-122511-20 entity=requests seq=10 key=None -> partition=1
12:25:11.383 [TEST] Waiting for the writers to process all messages...

Processing order for run nokey-122511 (20/20 processed):
  #  processed_at            writer  partition  entity    seq
  1  2026-11-02T12:25:11.684  A       1          pip       1
  2  2026-11-02T12:25:11.686  B       0          pip       2
  3  2026-11-02T12:25:12.010  A       1          lodash    1
  4  2026-11-02T12:25:12.010  B       0          pip       3
  5  2026-11-02T12:25:12.320  A       1          requests  1
  6  2026-11-02T12:25:12.320  B       0          lodash    2
  7  2026-11-02T12:25:12.630  A       1          requests  3
  8  2026-11-02T12:25:12.630  B       0          requests  2
  9  2026-11-02T12:25:12.939  A       1          lodash    5
 10  2026-11-02T12:25:12.939  B       0          requests  4
 11  2026-11-02T12:25:13.250  A       1          requests  9
 12  2026-11-02T12:25:13.250  B       0          lodash    3
 13  2026-11-02T12:25:13.560  A       1          requests  10
 14  2026-11-02T12:25:13.560  B       0          requests  5
 15  2026-11-02T12:25:13.868  B       0          lodash    4
 16  2026-11-02T12:25:14.180  B       0          requests  6
 17  2026-11-02T12:25:14.489  B       0          requests  7
 18  2026-11-02T12:25:14.798  B       0          pip       4
 19  2026-11-02T12:25:15.109  B       0          requests  8
 20  2026-11-02T12:25:15.422  B       0          lodash    6

Order check per package:
  lodash    partitions=['0', '1']  processed seq order=[1, 2, 5, 3, 4, 6]  -> OUT OF ORDER
  pip       partitions=['0', '1']  processed seq order=[1, 2, 3, 4]  -> IN ORDER
  requests  partitions=['0', '1']  processed seq order=[1, 3, 2, 4, 9, 10, 5, 6, 7, 8]  -> OUT OF ORDER
```

Ordering after fix



```
python .\writer.py A
12:25:11.690 [WRITER-A] COMMIT partition=1 offset=20 (next offset to read: 21)
12:25:11.693 [WRITER-A] RECEIVE from s0932_tasks partition=1 offset=21
12:25:11.694 [WRITER-A] correlation_id=nokey-122511-02
12:25:12.010 [WRITER-A] ORDER run=nokey-122511 entity=lodash seq=1 partition=1 offset=21
12:25:12.012 [WRITER-A] COMMIT partition=1 offset=21 (next offset to read: 22)
12:25:12.012 [WRITER-A] RECEIVE from s0932_tasks partition=1 offset=22
12:25:12.012 [WRITER-A] correlation_id=nokey-122511-05
12:25:12.321 [WRITER-A] ORDER run=nokey-122511 entity=requests seq=1 partition=1 offset=22
12:25:12.322 [WRITER-A] COMMIT partition=1 offset=22 (next offset to read: 23)
12:25:12.322 [WRITER-A] RECEIVE from s0932_tasks partition=1 offset=23
12:25:12.322 [WRITER-A] correlation_id=nokey-122511-08
12:25:12.630 [WRITER-A] ORDER run=nokey-122511 entity=requests seq=3 partition=1 offset=23
12:25:12.631 [WRITER-A] COMMIT partition=1 offset=23 (next offset to read: 24)
12:25:12.631 [WRITER-A] RECEIVE from s0932_tasks partition=1 offset=24
12:25:12.631 [WRITER-A] correlation_id=nokey-122511-13
12:25:12.939 [WRITER-A] ORDER run=nokey-122511 entity=lodash seq=5 partition=1 offset=24
12:25:12.941 [WRITER-A] COMMIT partition=1 offset=24 (next offset to read: 25)
12:25:12.941 [WRITER-A] RECEIVE from s0932_tasks partition=1 offset=25
12:25:12.941 [WRITER-A] correlation_id=nokey-122511-18
12:25:13.250 [WRITER-A] ORDER run=nokey-122511 entity=requests seq=9 partition=1 offset=25
12:25:13.252 [WRITER-A] COMMIT partition=1 offset=25 (next offset to read: 26)
12:25:13.252 [WRITER-A] RECEIVE from s0932_tasks partition=1 offset=26
12:25:13.252 [WRITER-A] correlation_id=nokey-122511-20
12:25:13.560 [WRITER-A] ORDER run=nokey-122511 entity=requests seq=10 partition=1 offset=26
12:25:13.562 [WRITER-A] COMMIT partition=1 offset=26 (next offset to read: 27)
12:28:13.212 [WRITER-A] RECEIVE from s0932_tasks partition=1 offset=27
12:28:13.213 [WRITER-A] correlation_id=key-122813-05
12:28:13.518 [WRITER-A] ORDER run=key-122813 entity=requests seq=1 partition=1 offset=27
12:28:13.520 [WRITER-A] COMMIT partition=1 offset=27 (next offset to read: 28)
12:28:13.521 [WRITER-A] RECEIVE from s0932_tasks partition=1 offset=28
12:28:13.521 [WRITER-A] correlation_id=key-122813-07
12:28:13.826 [WRITER-A] ORDER run=key-122813 entity=requests seq=2 partition=1 offset=28
12:28:13.828 [WRITER-A] COMMIT partition=1 offset=28 (next offset to read: 29)
12:28:13.828 [WRITER-A] RECEIVE from s0932_tasks partition=1 offset=29
12:28:13.828 [WRITER-A] correlation_id=key-122813-08
12:28:14.138 [WRITER-A] ORDER run=key-122813 entity=requests seq=3 partition=1 offset=29
12:28:14.139 [WRITER-A] COMMIT partition=1 offset=29 (next offset to read: 30)
12:28:14.140 [WRITER-A] RECEIVE from s0932_tasks partition=1 offset=30
12:28:14.140 [WRITER-A] correlation_id=key-122813-09
12:28:14.447 [WRITER-A] ORDER run=key-122813 entity=requests seq=4 partition=1 offset=30
12:28:14.449 [WRITER-A] COMMIT partition=1 offset=30 (next offset to read: 31)
12:28:14.450 [WRITER-A] RECEIVE from s0932_tasks partition=1 offset=31
12:28:14.450 [WRITER-A] correlation_id=key-122813-11
12:28:14.759 [WRITER-A] ORDER run=key-122813 entity=requests seq=5 partition=1 offset=31
12:28:14.760 [WRITER-A] COMMIT partition=1 offset=31 (next offset to read: 32)
12:28:14.760 [WRITER-A] RECEIVE from s0932_tasks partition=1 offset=32
12:28:14.760 [WRITER-A] correlation_id=key-122813-14
12:28:15.070 [WRITER-A] ORDER run=key-122813 entity=requests seq=6 partition=1 offset=32
12:28:15.072 [WRITER-A] COMMIT partition=1 offset=32 (next offset to read: 33)
12:28:15.072 [WRITER-A] RECEIVE from s0932_tasks partition=1 offset=33
12:28:15.072 [WRITER-A] correlation_id=key-122813-15
12:28:15.382 [WRITER-A] ORDER run=key-122813 entity=requests seq=7 partition=1 offset=33
12:28:15.384 [WRITER-A] COMMIT partition=1 offset=33 (next offset to read: 34)
12:28:15.384 [WRITER-A] RECEIVE from s0932_tasks partition=1 offset=34
12:28:15.384 [WRITER-A] correlation_id=key-122813-17
12:28:15.695 [WRITER-A] ORDER run=key-122813 entity=requests seq=8 partition=1 offset=34
12:28:15.697 [WRITER-A] COMMIT partition=1 offset=34 (next offset to read: 35)
12:28:15.697 [WRITER-A] RECEIVE from s0932_tasks partition=1 offset=35
12:28:15.697 [WRITER-A] correlation_id=key-122813-18
12:28:16.005 [WRITER-A] ORDER run=key-122813 entity=requests seq=9 partition=1 offset=35
12:28:16.007 [WRITER-A] COMMIT partition=1 offset=35 (next offset to read: 36)
12:28:16.007 [WRITER-A] RECEIVE from s0932_tasks partition=1 offset=36
12:28:16.007 [WRITER-A] correlation_id=key-122813-20
12:28:16.318 [WRITER-A] ORDER run=key-122813 entity=requests seq=10 partition=1 offset=36
12:28:16.319 [WRITER-A] COMMIT partition=1 offset=36 (next offset to read: 37)
```

```
PS C:\Users\natal\Documents\masters\Fall 26\data 260\github\data260-0932\reports\hw08\agents
ython .\writer.py B
12:25:13.562 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=42
12:25:13.562 [WRITER-B] correlation_id=nokey-122511-12
12:25:13.868 [WRITER-B] ORDER run=nokey-122511 entity=lodash seq=4 partition=0 offset=42
12:25:13.869 [WRITER-B] COMMIT partition=0 offset=42 (next offset to read: 43)
12:25:13.869 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=43
12:25:13.869 [WRITER-B] correlation_id=nokey-122511-14
12:25:14.181 [WRITER-B] ORDER run=nokey-122511 entity=requests seq=6 partition=0 offset=43
12:25:14.182 [WRITER-B] COMMIT partition=0 offset=43 (next offset to read: 44)
12:25:14.182 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=44
12:25:14.182 [WRITER-B] correlation_id=nokey-122511-15
12:25:14.489 [WRITER-B] ORDER run=nokey-122511 entity=requests seq=7 partition=0 offset=44
12:25:14.491 [WRITER-B] COMMIT partition=0 offset=44 (next offset to read: 45)
12:25:14.491 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=45
12:25:14.491 [WRITER-B] correlation_id=nokey-122511-16
12:25:14.798 [WRITER-B] ORDER run=nokey-122511 entity=pip seq=4 partition=0 offset=45
12:25:14.800 [WRITER-B] COMMIT partition=0 offset=45 (next offset to read: 46)
12:25:14.800 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=46
12:25:14.800 [WRITER-B] correlation_id=nokey-122511-17
12:25:15.110 [WRITER-B] ORDER run=nokey-122511 entity=requests seq=8 partition=0 offset=46
12:25:15.111 [WRITER-B] COMMIT partition=0 offset=46 (next offset to read: 47)
12:25:15.111 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=47
12:25:15.111 [WRITER-B] correlation_id=nokey-122511-19
12:25:15.422 [WRITER-B] ORDER run=nokey-122511 entity=lodash seq=6 partition=0 offset=47
12:25:15.423 [WRITER-B] COMMIT partition=0 offset=47 (next offset to read: 48)
12:28:13.209 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=48
12:28:13.209 [WRITER-B] correlation_id=key-122813-01
12:28:13.518 [WRITER-B] ORDER run=key-122813 entity=pip seq=1 partition=0 offset=48
12:28:13.520 [WRITER-B] COMMIT partition=0 offset=48 (next offset to read: 49)
12:28:13.521 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=49
12:28:13.521 [WRITER-B] correlation_id=key-122813-02
12:28:13.826 [WRITER-B] ORDER run=key-122813 entity=lodash seq=1 partition=0 offset=49
12:28:13.828 [WRITER-B] COMMIT partition=0 offset=49 (next offset to read: 50)
12:28:13.828 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=50
12:28:13.828 [WRITER-B] correlation_id=key-122813-03
12:28:14.138 [WRITER-B] ORDER run=key-122813 entity=pip seq=2 partition=0 offset=50
12:28:14.139 [WRITER-B] COMMIT partition=0 offset=50 (next offset to read: 51)
12:28:14.140 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=51
12:28:14.140 [WRITER-B] correlation_id=key-122813-04
12:28:14.447 [WRITER-B] ORDER run=key-122813 entity=pip seq=3 partition=0 offset=51
12:28:14.449 [WRITER-B] COMMIT partition=0 offset=51 (next offset to read: 52)
12:28:14.449 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=52
12:28:14.450 [WRITER-B] correlation_id=key-122813-06
12:28:14.758 [WRITER-B] ORDER run=key-122813 entity=lodash seq=2 partition=0 offset=52
12:28:14.760 [WRITER-B] COMMIT partition=0 offset=52 (next offset to read: 53)
12:28:14.760 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=53
12:28:14.760 [WRITER-B] correlation_id=key-122813-10
12:28:15.070 [WRITER-B] ORDER run=key-122813 entity=lodash seq=3 partition=0 offset=53
12:28:15.072 [WRITER-B] COMMIT partition=0 offset=53 (next offset to read: 54)
12:28:15.072 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=54
12:28:15.072 [WRITER-B] correlation_id=key-122813-12
12:28:15.382 [WRITER-B] ORDER run=key-122813 entity=lodash seq=4 partition=0 offset=54
12:28:15.384 [WRITER-B] COMMIT partition=0 offset=54 (next offset to read: 55)
12:28:15.384 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=55
12:28:15.384 [WRITER-B] correlation_id=key-122813-13
12:28:15.695 [WRITER-B] ORDER run=key-122813 entity=lodash seq=5 partition=0 offset=55
12:28:15.697 [WRITER-B] COMMIT partition=0 offset=55 (next offset to read: 56)
12:28:15.697 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=56
12:28:15.697 [WRITER-B] correlation_id=key-122813-16
12:28:16.005 [WRITER-B] ORDER run=key-122813 entity=pip seq=4 partition=0 offset=56
12:28:16.007 [WRITER-B] COMMIT partition=0 offset=56 (next offset to read: 57)
12:28:16.007 [WRITER-B] RECEIVE from s0932_tasks partition=0 offset=57
12:28:16.007 [WRITER-B] correlation_id=key-122813-19
12:28:16.318 [WRITER-B] ORDER run=key-122813 entity=lodash seq=6 partition=0 offset=57
12:28:16.319 [WRITER-B] COMMIT partition=0 offset=57 (next offset to read: 58)
```

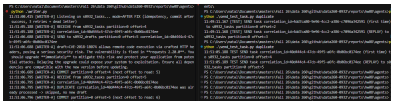
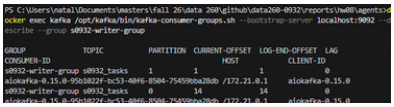
```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw08\agents> python .\ordering_test.py key
  12:28:13.192 [TEST] run=key-122813 VERIFY_SEED=260932 messages=20
  12:28:13.208 [TEST] SEND key-122813-01 entity=pip seq=1 key=b'pip' -> partition=0
  12:28:13.210 [TEST] SEND key-122813-02 entity=lodash seq=1 key=b'lodash' -> partition=0
  12:28:13.210 [TEST] SEND key-122813-03 entity=pip seq=2 key=b'pip' -> partition=0
  12:28:13.211 [TEST] SEND key-122813-04 entity=pip seq=3 key=b'pip' -> partition=0
  12:28:13.212 [TEST] SEND key-122813-05 entity=requests seq=1 key=b'requests' -> partition=1
  12:28:13.213 [TEST] SEND key-122813-06 entity=lodash seq=2 key=b'lodash' -> partition=0
  12:28:13.213 [TEST] SEND key-122813-07 entity=requests seq=2 key=b'requests' -> partition=1
  12:28:13.214 [TEST] SEND key-122813-08 entity=requests seq=3 key=b'requests' -> partition=1
  12:28:13.215 [TEST] SEND key-122813-09 entity=requests seq=4 key=b'requests' -> partition=1
  12:28:13.216 [TEST] SEND key-122813-10 entity=lodash seq=3 key=b'lodash' -> partition=0
  12:28:13.217 [TEST] SEND key-122813-11 entity=requests seq=5 key=b'requests' -> partition=1
  12:28:13.217 [TEST] SEND key-122813-12 entity=lodash seq=4 key=b'lodash' -> partition=0
  12:28:13.218 [TEST] SEND key-122813-13 entity=lodash seq=5 key=b'lodash' -> partition=0
  12:28:13.219 [TEST] SEND key-122813-14 entity=requests seq=6 key=b'requests' -> partition=1
  12:28:13.220 [TEST] SEND key-122813-15 entity=requests seq=7 key=b'requests' -> partition=1
  12:28:13.221 [TEST] SEND key-122813-16 entity=pip seq=4 key=b'pip' -> partition=0
  12:28:13.221 [TEST] SEND key-122813-17 entity=requests seq=8 key=b'requests' -> partition=1
  12:28:13.222 [TEST] SEND key-122813-18 entity=requests seq=9 key=b'requests' -> partition=1
  12:28:13.223 [TEST] SEND key-122813-19 entity=lodash seq=6 key=b'lodash' -> partition=0
  12:28:13.224 [TEST] SEND key-122813-20 entity=requests seq=10 key=b'requests' -> partition=1
  12:28:13.224 [TEST] Waiting for the writers to process all messages...

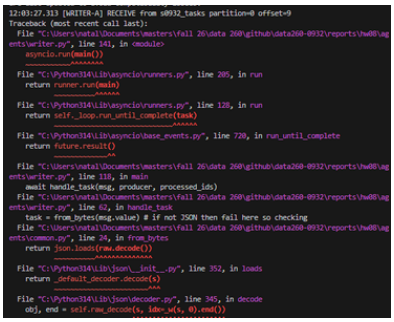
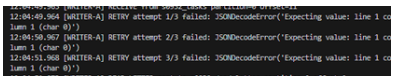
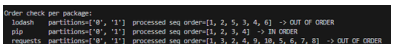
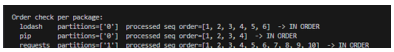
  Processing order for run key-122813 (20/20 processed):
    #  processed_at            writer  partition  entity    seq
    1  2026-11-02T12:28:13.517  A       1          requests  1
    2  2026-11-02T12:28:13.517  B       0          pip       1
    3  2026-11-02T12:28:13.826  A       1          requests  2
    4  2026-11-02T12:28:13.826  B       0          lodash    1
    5  2026-11-02T12:28:14.138  A       1          requests  3
    6  2026-11-02T12:28:14.138  B       0          pip       2
    7  2026-11-02T12:28:14.447  A       1          requests  4
    8  2026-11-02T12:28:14.447  B       0          pip       3
    9  2026-11-02T12:28:14.758  A       1          requests  5
   10  2026-11-02T12:28:14.758  B       0          lodash    2
   11  2026-11-02T12:28:15.070  A       1          requests  6
   12  2026-11-02T12:28:15.070  B       0          lodash    3
   13  2026-11-02T12:28:15.382  A       1          requests  7
   14  2026-11-02T12:28:15.382  B       0          lodash    4
   15  2026-11-02T12:28:15.695  A       1          requests  8
   16  2026-11-02T12:28:15.695  B       0          lodash    5
   17  2026-11-02T12:28:16.005  A       1          requests  9
   18  2026-11-02T12:28:16.005  B       0          pip       4
   19  2026-11-02T12:28:16.317  A       1          requests  10
   20  2026-11-02T12:28:16.317  B       0          lodash    6

  Order check per package:
    lodash    partitions=['0']  processed seq order=[1, 2, 3, 4, 5, 6]  -> IN ORDER
    pip       partitions=['0']  processed seq order=[1, 2, 3, 4]  -> IN ORDER
    requests  partitions=['1']  processed seq order=[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]  -> IN ORDER
```

V. Comparison

The completed failure table, the matching log excerpt for every row is fully shown in the above sections for visibility (i included some important small parts of the full logs for this table)

Failure scenario	Before fix	After fix
Duplicate delivery		
Consumer killed mid-work		

Poison message		
Two partitions/two consumers		

Idempotency

```python
def load_processed_ids():
    if not os.path.exists(PROCESSED_FILE):
        return set()
    with open(PROCESSED_FILE) as f:
        return set(line.strip() for line in f if line.strip())

def remember_processed_id(correlation_id, processed_ids):
    processed_ids.add(correlation_id)
    with open(PROCESSED_FILE, "a") as f:
        f.write(correlation_id + "\n")
```

```python
    #idempotency check
    if not BEFORE_FIX and correlation_id in processed_ids:
        log(f"{TAG} DUPLICATE correlation_id={correlation_id} was already processed -> skipped, no new draft")
        return
```

Offset-commit

```python
        consumer = AIOKafkaConsumer(
            TASKS_TOPIC,
            bootstrap_servers=KAFKA_BOOTSTRAP_SERVERS,
            group_id="s0932-writer-group",
            auto_offset_reset="earliest",
            enable_auto_commit=BEFORE_FIX,
        )
```

```python
            await consumer.commit()
            log(f"{TAG} COMMIT partition={msg.partition} offset={msg.offset} (next offset to read: {msg.offset + 1})")
```

retry/dead-letter

```python
            #retries then send to dead letter topic
            for attempt in range(1, MAX_RETRIES + 1):
                try:
                    await handle_task(msg, producer, processed_ids)
                    break
                except Exception as error:
                    log(f"{TAG} RETRY attempt {attempt}/{MAX_RETRIES} failed: {error!r}")
                    if attempt == MAX_RETRIES:
                        await send_to_dead_letter(producer, msg, error)
                    else:
                        await asyncio.sleep(1)
```

```python
async def send_to_dead_letter(producer, msg, error):
    dead_letter = {
        "source_topic": msg.topic,
        "partition": msg.partition,
        "offset": msg.offset,
        "original_message": msg.value.decode(errors="replace"),
        "error": repr(error),
        "attempts": MAX_RETRIES,
    }
    metadata = await producer.send_and_wait(DEAD_LETTER_TOPIC, to_bytes(dead_letter))
    log(f"{TAG} DEAD LETTER sent to {metadata.topic} partition={metadata.partition} offset={metadata.offset}")
    log(f"{TAG} DEAD LETTER message={dead_letter}")
```

partition-key code

```python
        # Run 2 uses the package name as the key, so one package always goes to one partition
        key = test["entity"].encode() if mode == "key" else None
        metadata = await producer.send_and_wait(TASKS_TOPIC, to_bytes(message), key=key)
        log(f"[TEST] SEND {message['correlation_id']} entity={test['entity']} seq={test['seq']} key={key} -> partition={metadata.partition}")
```

Your two-benefits/two-responsibilities comparison with the HW2 in-process graph.

If one node crashed, the whole program stopped and the state was lost for homework 2. Kafka has its own process. When I killed the writer in Part 3, the planner and reviewer was able to keep running. The unfinished message waited in s0932_tasks until the writer came back. In HW2, there was only one copy of each node. With Kafka, I started two writers in the same consumer group. Kafka split the two partitions of s0932_tasks between them so that way the writers can work in parallel without any change to the planner or reviewer.

HW2 has each function called exactly once. Kafka has the same message and can be delivered more than once. Therefore, I added idempotency checks. The writer remembers each correlation_id and skips replays so it handles duplicates. In HW2, the steps always ran in the order the graph defined. However, Kafka makes it so messages on different partitions can be processed out of order by different consumers. I choose a package name in order for related messages to stay in one partition and keep their order. Therefore ordering is a responsibility.

AI_USE.md

reports > hw08 > AI_USE.md

1. I used AI for looking for docker commands for the logs for part 1 and 3 in order to show what was needed. For example, the lag command to make sure it was showing correctly. I also used is so that I could run the same writer file for different parts of the homework instead of making different files which is what I have before. I made the files by myself first and used the lecture and demo as references. I then used AI to help me with different docker commands and figuring out the docker files in my github report because I was unsure how to do that part. I also used it to help debug when I was unable to run all 20 runs for part 3.
2. With AI help I created one csv file output for the 20 runs but I would it was wrong because it seemed that the two writers were overwritting eachother. I checked it manually and found it was this case. So, I seperated into two csv files one for each writer. I also used AI to help debug the error and it also conculded the same thing that was causing it to only run 12 out of the 20 runs.
3. I detected this problem by looking at the output of running the python file when it said it was in order, which is what I want, but it did not go through all 20 files. I also checked the one where it wasnt it order and found out that the out of order should appear one more time. So, with this caught, I looked into rewriting the writer function in my coding files to help fix the error.
4. I rewrote the writer.py file and ordering_test.py file so that all 20 runs are processed. The writers wrote to the same file but because I saw that the logs of the writers finished around the same time then they overworte eachother. Thereofre, the rows got lost and the runs werent done completely. With the writer having its own file, I was able to verify all 20 runs are done correctly. I think I could also try timing the second writer to start that a while later than the first one. However, that seemed to not really affect the runs as I still missed some runs.